Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering[1]
Student Name	Amilineni Gopika
Register Number	3122247001003
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	21/09/2026

1 Objective

Try clustering on the Human Activity Recognition dataset three different ways – K-Means, DBSCAN and Hierarchical Agglomerative Clustering – and actually see how each one carves up the data compared to the six real activity labels, instead of just assuming clustering will neatly rediscover the activities on its own.

2 Problem Statement

Given the UCI HAR smartphone sensor dataset, preprocess and standardize it, reduce it for visualization, then run K-Means (choosing k with the elbow method), DBSCAN (tuning ϵ and minPts), and Agglomerative Clustering (comparing linkage methods) on it. Compare the three using internal metrics (silhouette, Davies-Bouldin, Calinski-Harabasz) and external metrics against the true activity labels (ARI, NMI), and figure out which algorithm actually produces the most meaningful clusters here.

3 Dataset Description

Dataset	Source	Samples	Features	Activities	Subjects
Human Activity Recognition Using Smartphones	UCI Machine Learning Repository	10299 (7352 train + 2947 test)	561	6	30

Activity distribution: LAYING = 1944, STANDING = 1906, SITTING = 1777, WALKING = 1722, WALKING_UPSTAIRS = 1544, WALKING_DOWNSTAIRS = 1406. Reasonably balanced across the six classes, with LAYING slightly over-represented and WALKING_DOWNSTAIRS slightly under.

Preprocessing steps:

- Merged the official train and test splits back into one 10299-row dataset, since clustering is unsupervised and doesn't need the original train/test split, and combining the two just gives a fuller picture of the activity space.
- No missing values anywhere in the 561 features, so no imputation was needed.

[1] **GitHub Repository:** https://github.com/sanjay-6727/Introdution-to-Machine-Learning-

- Activity codes (1–6) mapped to their proper names using `activity_labels.txt`, kept around purely as ground truth for the external evaluation metrics, never fed to the clustering algorithms themselves.
- All 561 features standardized with `StandardScaler`, since clustering algorithms are distance-based and the raw features already sit in roughly $[-1, 1]$ but on slightly different scales.
- Since 10299 points is a lot to run DBSCAN, Agglomerative clustering and t-SNE on comfortably, I took a stratified 2000-point subsample (proportional per activity) for the actual clustering and visualization, so every algorithm sees the same data and the comparison stays fair.
- PCA down to 30 components was used as the working feature space for all three clustering algorithms (about 82% of the variance retained), mainly to keep distances meaningful and DBSCAN/HAC tractable; a separate 2-component PCA and a 2D t-SNE embedding were computed purely for plotting.

4 Model A: K-Means Clustering and the Elbow Method

Table 2: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	433577.6	0.4775
3	359760.2	0.4070
4	332567.7	0.2174
5	305593.0	0.1940
6	291478.2	0.1835
7	280425.3	0.1602
8	270008.5	0.1573

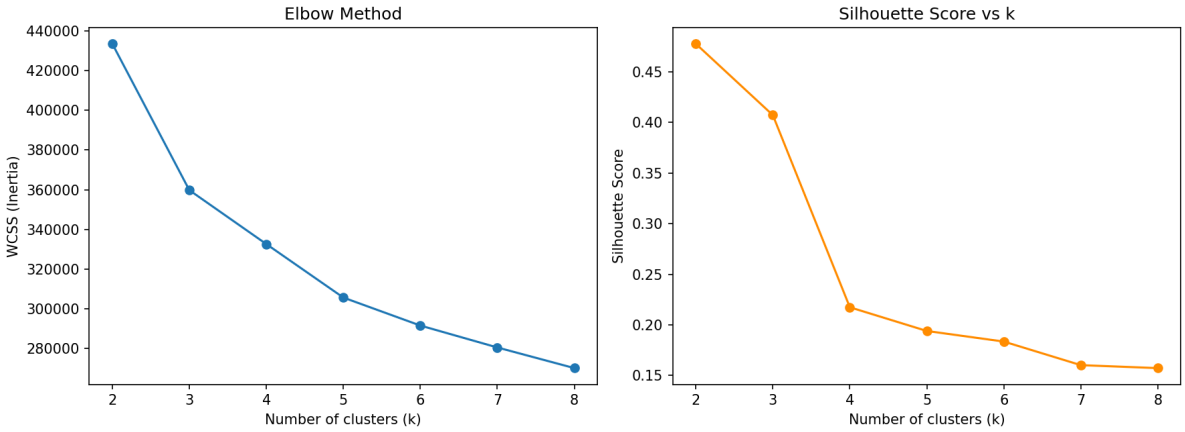


Figure 1: Elbow curve (WCSS vs k) and silhouette score vs k for K-Means.

Purely on silhouette score, $k = 2$ wins by a wide margin, and the elbow curve also bends hardest around $k = 2$ to $k = 3$. That's not surprising – at the coarsest level, HAR activities split into two very distinct physical states, moving (the three walking types) and stationary (sitting, standing, laying), and geometrically those two blobs are far apart while the finer distinctions inside each blob are much subtler. Since the assignment goal is to compare clustering against the six known activities, though, I used **k=6** for the main analysis and all cross-algorithm comparisons, so K-Means, DBSCAN and Hierarchical Clustering are all being judged on the same basis.

5 Model B: DBSCAN

minPts was fixed at 60 (twice the 30-dimensional PCA space, a common rule of thumb), and ϵ was swept using a k-distance plot plus a direct search over several candidate values.

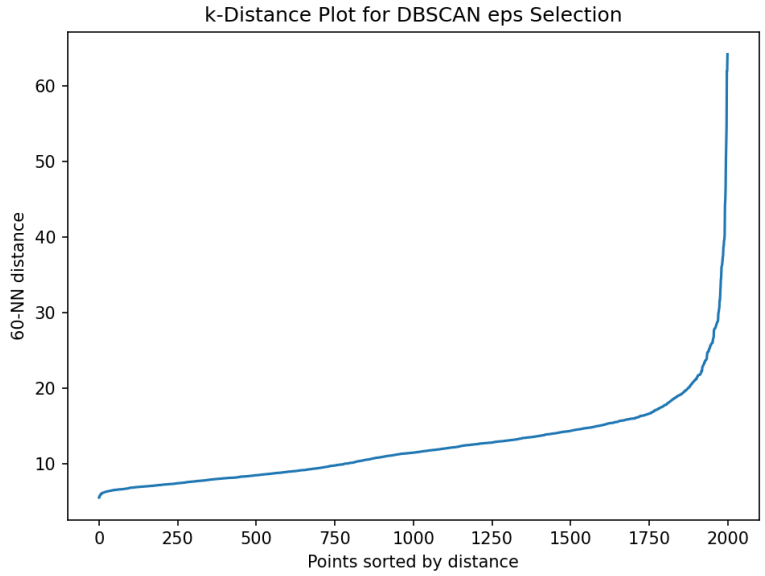


Figure 2: 60-NN distance plot used to guide the choice of ϵ for DBSCAN.

Table 3: DBSCAN – eps Tuning

eps	Clusters Found	Noise Points	Silhouette (non-noise)
8	1	1220 (61.0%)	–
10	1	996 (49.8%)	–
12	2	483 (24.1%)	0.5011
14	1	219 (11.0%)	–
16	1	108 (5.4%)	–
18	1	70 (3.5%)	–
20	1	37 (1.9%)	–

$\epsilon = 12$ was the only setting in the sweep that actually produced more than one cluster; every other value either merged everything into one blob (too permissive) or scattered most points as noise (too strict). The final DBSCAN run at $\epsilon = 12$, minPts $= 60$ gives **2 clusters** and 483 noise points (24.1% of the data).

6 Model C: Hierarchical Agglomerative Clustering

Table 4: Hierarchical Clustering – Linkage Comparison (k=6)

Linkage	Silhouette Score	Davies-Bouldin Index
Single	0.5590	0.4092
Complete	0.4458	1.0389
Average	0.4625	0.9031
Ward	0.1663	1.8512

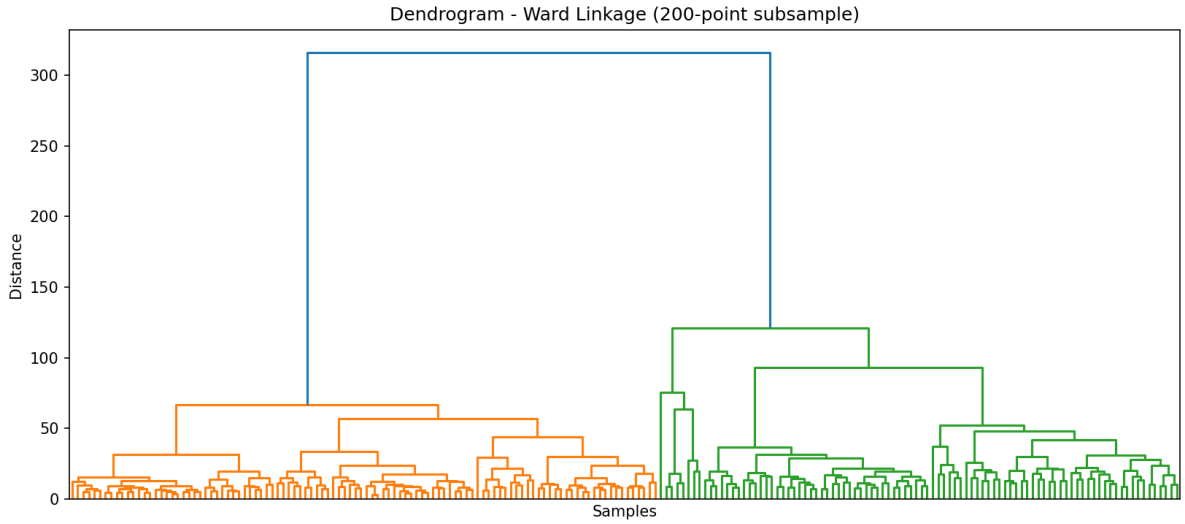


Figure 3: Dendrogram using Ward's linkage on a 200-point subsample.

Single linkage posts the best silhouette score here, but that's a classic chaining artifact – it tends to swallow almost everything into one giant cluster and peel off a handful of outliers as tiny separate clusters, which looks great by silhouette but doesn't map onto anything meaningful. Ward's linkage, which the experiment specifically asks for, gives a much lower silhouette (0.1663) but that's because it's actually trying to build six balanced, compact clusters instead of one big blob plus scraps, so it's the more honest and more useful result even though the raw silhouette number looks worse on paper. Ward's linkage is what's used for the rest of the comparison below.

7 Evaluation Metrics and Results

Table 5: Clustering Evaluation Metrics Comparison

Model	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means (k=6)	0.1835	1.7524	858.3	0.4178	0.5629
DBSCAN (eps=12)	0.5011	0.8295	1810.5	0.3022	0.5321
Hierarchical (Ward)	0.1663	1.8512	794.6	0.3541	0.5345

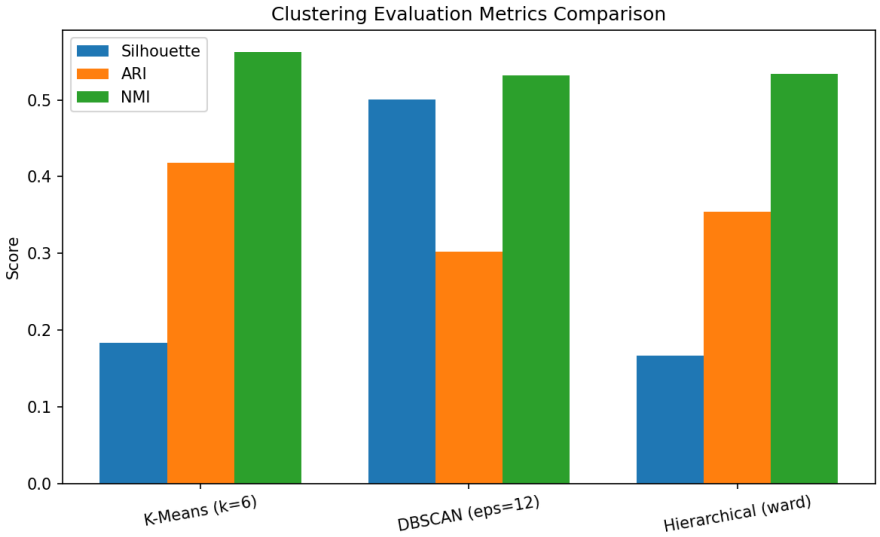


Figure 4: Silhouette, ARI and NMI compared across the three clustering algorithms.

K-Means comes out ahead on the metrics that actually compare against the true activity labels (ARI 0.4178, NMI 0.5629), even though DBSCAN wins on the purely internal silhouette and Davies-Bouldin scores. That's the DBSCAN result being a bit of a mirage: it only found 2 clusters (moving vs stationary), and it's easy to get a great internal separation score with just 2 well-separated blobs while completely missing the finer 6-way structure the experiment actually cares about.

8 Graphs and Visualization

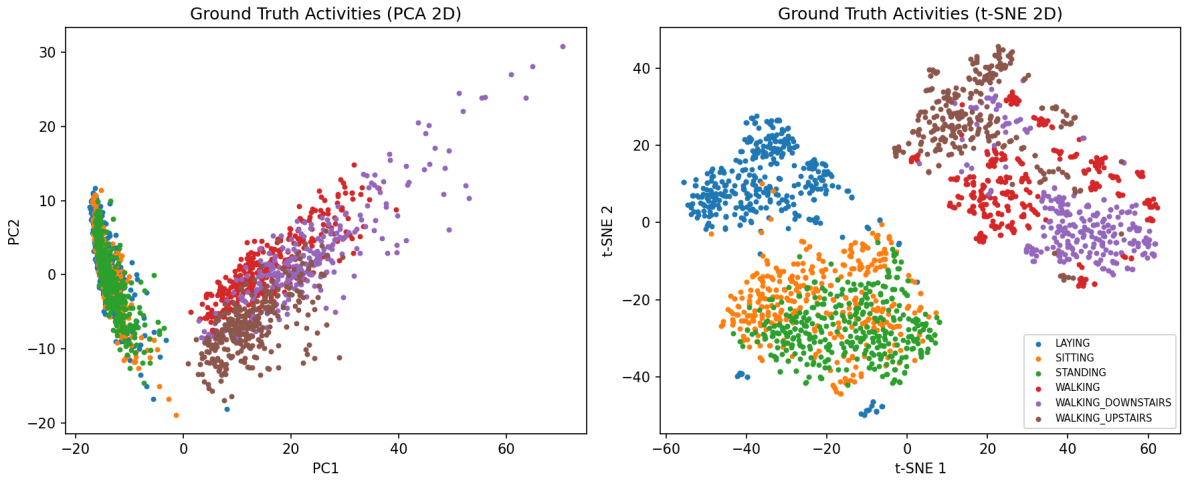


Figure 5: PCA (left) and t-SNE (right) 2D projections colored by the true activity label, for reference.

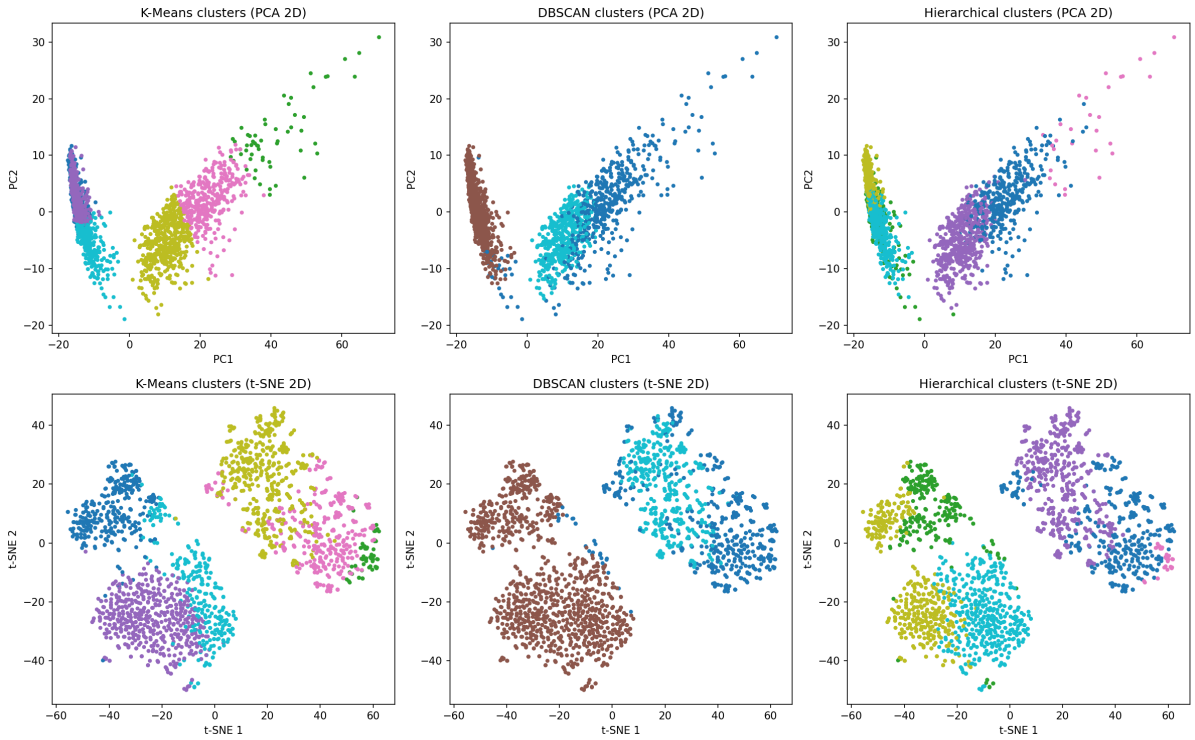


Figure 6: PCA (top row) and t-SNE (bottom row) projections colored by cluster assignment for K-Means, DBSCAN and Hierarchical (Ward) respectively.



Figure 7: K-Means clusters (k=6) cross-tabulated against the true activity labels.

Inference: The ground-truth t-SNE plot shows a very clean, tight LAYING blob sitting far from everything else, a fairly clean WALKING/WALKING_UPSTAIRS/WALKING_DOWNSTAIRS

region, and SITTING and STANDING overlapping quite a bit with each other. That overlap between SITTING and STANDING is exactly where all three clustering algorithms struggle: in the K-Means cross-tab, cluster 2 mixes 244 SITTING with 249 STANDING points almost 50/50, and cluster 5 mixes 88 SITTING, 121 STANDING and 56 LAYING together. LAYING, by contrast, gets almost cleanly isolated into cluster 0 (305 of 378 LAYING points), and the three walking activities mostly separate out into clusters 1, 3 and 4, though WALKING_UPSTAIRS and WALKING_DOWNSTAIRS bleed into each other somewhat too.

9 Observations

- **Which algorithm produced the most meaningful clusters:** K-Means, on balance. It's the only one of the three that actually attempts and roughly recovers something close to the real 6-way activity structure (ARI 0.418, NMI 0.563), even if it's far from perfect. DBSCAN's headline internal scores are misleading once you check them against ARI/NMI – it settled for a coarse 2-cluster solution that trades away the fine-grained structure entirely.

- **How sensitive was K-Means to the choice of k:** Very. The silhouette score drops sharply from 0.4775 at k=2 to 0.1835 at k=6, meaning the natural, cleanly-separated structure in this data is really just 2 clusters (moving vs stationary); forcing k=6 finds real sub-structure but at a real cost to how cleanly separated the individual clusters are.

- **Did DBSCAN detect noise or small clusters effectively:** It found noise, plenty of it, but not small clusters – every ϵ that avoided lumping everything into one cluster still only produced 2 clusters, never anything close to 6. It's better described here as a noise/outlier detector than as a way to recover the six activities.

- **How does linkage choice affect Hierarchical Clustering:** Hugely. Single linkage chains points together and basically produces one dominant cluster plus a scattering of near-singleton outlier clusters, which inflates the silhouette score without meaning much. Complete and average linkage sit in between. Ward's linkage is the only one that produces genuinely balanced, compact clusters at k=6, which is presumably why the experiment specifically calls for it, even though its raw silhouette number is the lowest of the four.

- **Which internal metric best matched visual intuition:** Calinski-Harabasz tracked the t-SNE/PCA plots better than silhouette did in this case – K-Means' relatively high CH score (858.3) lines up with clusters that look reasonably compact and separated in the scatter plots, whereas silhouette got fooled by DBSCAN's 2-cluster solution looking artificially clean.

10 Discussion

The big theme here is that "looks good by an internal metric" and "actually recovers the known structure" are not the same thing, and DBSCAN is the clearest example: best silhouette and Davies-Bouldin scores of the three, but worst ARI, because it quietly gave

up on separating six activities and settled for two. K-Means, despite having the worst silhouette score of the three at k=6, is the one that comes closest to matching the ground truth, simply because it was forced to actually produce six clusters rather than finding an easy two-way split. The SITTING/STANDING overlap that trips up every algorithm here isn't really a clustering failure so much as a property of the data itself – those two activities produce genuinely similar sensor signatures when a phone is worn at the waist, so no amount of retuning ϵ or trying a different linkage is likely to fully separate them without extra features or a different sensor placement.

11 Conclusion

Across K-Means, DBSCAN and Hierarchical (Ward) clustering on the HAR dataset, K-Means at $k = 6$ gave the most useful match to the real six activities (ARI 0.418, NMI 0.563), mainly by being willing to accept a lower silhouette score in exchange for actually splitting the data six ways. DBSCAN is genuinely good at finding the coarsest possible structure (moving vs. stationary) and flagging outliers, but it isn't well suited to recovering finer-grained classes on this dataset. Ward's linkage HAC lands in between the two, doing reasonably on external metrics without either of K-Means' flexibility or DBSCAN's noise handling. If the actual goal is to recover activity-level clusters, K-Means with a task-informed choice of k (rather than the value silhouette alone would suggest) is the more practical choice here.

12 Learning Outcomes

- Learned to read an elbow curve and a silhouette-vs-k plot together, and to recognize when they disagree with what the actual task needs (silhouette wanted $k = 2$, the task wanted $k = 6$).
- Got hands-on practice tuning DBSCAN's ϵ using a k-distance plot, and saw first-hand how narrow the useful range of ϵ can be on high-dimensional, PCA-reduced sensor data.
- Directly observed the single-linkage chaining problem in hierarchical clustering, and understood why Ward's linkage is usually preferred for producing balanced clusters even when its raw silhouette looks worse.
- Practiced distinguishing internal clustering metrics (silhouette, Davies-Bouldin, Calinski-Harabasz) from external ones (ARI, NMI), and saw a concrete case where they disagreed.
- Used PCA and t-SNE together for visualization, and noticed how t-SNE separates the SITTING/STANDING overlap a little better than a straight 2D PCA projection does.

13 References

1. Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra and Jorge L. Reyes-Ortiz, "Human Activity Recognition on Smartphones using a Multiclass Hardware-

Friendly Support Vector Machine", IWAAL 2012.

2. UCI Machine Learning Repository, Human Activity Recognition Using Smartphones Data Set: https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones

3. Scikit-learn documentation: https://scikit-learn.org/stable/modules/clustering.html